# Huawei MG-ACC-46 Access Node Hardware/Environmental Fault Resolution SOP (Major Severity)

**Node:** MG-ACC-46 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_2902

## Description

The Huawei MG-ACC-46 Access Node failure in the MG Road Core site is attributed to Hardware/Environmental Faults, specifically ASIC memory parity errors, line card crashes, and power supply redundancy loss. This failure domain occurs due to the inherent design of the Access Node's distributed architecture, where multiple line cards and power supplies are deployed in a redundant configuration to ensure high availability. However, this redundancy can be compromised by a combination of hardware and environmental factors, such as power supply unit (PSU) failures, temperature fluctuations, and humidity levels that exceed the recommended operating range. Furthermore, the ASIC memory parity errors can be triggered by a variety of factors, including software bugs, hardware defects, or electromagnetic interference (EMI). In addition, the line card crashes can be caused by issues with the line card's firmware, software, or hardware components. As a result, the Access Node's control plane and data plane may become unstable, leading to a major severity alarm (ALARM_2902) and impacting the overall network performance and reliability.

## Pre-requisite Checks

- task: Verify the site location and ensure that remote hands are dispatched to the MG Road Core site., description: Confirm that the remote hands have access to the site and are equipped with the necessary tools and equipment to perform the required tasks.

- task: Check the redundant link status to ensure that the network is not impacted by the Access Node failure., description: Verify that the redundant links are up and functioning properly to ensure that the network can maintain its availability and performance.

- task: Review the system logs to identify any potential issues or patterns that may have led to the failure., description: Analyze the logs to determine if there were any software or hardware issues that may have contributed to the failure.

## Potential Root Causes

| Cause | Probability |
|---|---|
| ASIC memory parity error caused by software bug or hardware defect | High |
| Line card crash due to firmware or software issue | Medium |
| Power supply redundancy loss due to PSU failure or environmental conditions | Medium |
| Electromagnetic interference (EMI) affecting the ASIC memory or line card | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display system internal, Expected Output: System Internal Information, Description: This command will display the system internal information, including the status of the ASIC memory, line cards, and power supplies. The output will help identify if there are any issues with the hardware components.

Command: display fault-management fault, Expected Output: Fault Management Information, Description: This command will display the fault management information, including the fault type, location, and timestamp. The output will help identify the root cause of the failure and the severity level.

Command: display system log, Expected Output: System Log Information, Description: This command will display the system log information, including the system events, errors, and warnings. The output will help identify if there were any software or hardware issues that may have contributed to the failure.
```

## Resolution Steps

Step 1: Step: Replace the faulty ASIC or line card, Command: replace slot , Description: Replace the faulty ASIC or line card with a new one to restore the system's functionality. Ensure that the replacement is done by a trained technician to avoid any potential issues.

Step 2: Step: Replace the faulty power supply unit (PSU), Command: replace psu , Description: Replace the faulty PSU with a new one to restore the system's power supply redundancy. Ensure that the replacement is done by a trained technician to avoid any potential issues.

Step 3: Step: Update the system software or firmware, Command: upload software , Description: Update the system software or firmware to the latest version to ensure that any potential issues are resolved. Ensure that the update is done by a trained technician to avoid any potential issues.

## Post-Check Verification

```
Command: display system internal, Expected Output: System Internal Information, Description: Verify that the system internal information is displayed correctly and that there are no issues with the hardware components.

Command: display fault-management fault, Expected Output: Fault Management Information, Description: Verify that the fault management information is displayed correctly and that the fault has been cleared.

Command: display system log, Expected Output: System Log Information, Description: Verify that the system log information is displayed correctly and that there are no errors or warnings related to the failure.
```

## Roll-back Procedures

- Step: Rollback the system software or firmware update, Command: rollback software , Description: Rollback the system software or firmware update to the previous version in case the fix fails.

- Step: Replace the replaced hardware component, Command: replace slot , Description: Replace the replaced hardware component with the original one in case the fix fails.

- Step: Review the system logs and configuration files, Command: display system log, Description: Review the system logs and configuration files to ensure that the fix did not introduce any new issues.